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PACKAGING CONTAINER AND SOFT FOOD-CONTAINED PACKAGE

Abstract

A packaging container, in which a soft food and a liquid are contained and which is sealed by a cover, includes a receptacle portion in which a recess portion, in which the soft food is contained, and a head space that is filled with the liquid above the soft food, are formed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a Continuation Application of PCT Application No. PCT/JP2023/041865, filed Nov. 21, 2023 and based upon and claiming the benefit of priority from Japanese Patent Application No. 2022-185813, filed Nov. 21, 2022, the entire contents of all of which are incorporated herein by reference.

FIELD

[0002] The present invention relates to a packaging container in which a soft food is contained, and a soft food-contained package in which a soft food is contained in the packaging container.

BACKGROUND

[0003] In recent years, soft foods are distributed as foods for nursing care for the elderly with difficulty in chewing or swallowing. A soft food is, for example, a food produced by griding a solid food and then solidifying and re-forming the resultant in a desired shape. In addition, as a soft food-contained package, as disclosed in International Publication No. 2013/146618, there is known a product in which a mousse-like food as a soft food, and a seasoning liquid as a liquid, are filled in a cup container that is a packaging container, and the cup container is sealed by a seal cover.

SUMMARY

[0004] According to one aspect of the present invention, a packaging container is a packaging container in which a soft food and a liquid are contained and which is sealed by a cover, the packaging container including a receptacle portion in which a recess portion, in which the soft food is contained, and a head space that is filled with the liquid above the soft food, are formed.

[0005] According to another aspect of the present invention, there are included a soft food, a liquid, a receptacle portion in which a recess portion, in which the soft food is contained, and a head space that is filled with the liquid above the soft food, are formed, and a cover sealing the receptacle portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a side view illustrating, by a partial cross section, a structure of a soft food-contained package according to an embodiment of the present invention.

[0007] FIG. 2 is a plan view illustrating a structure of a packaging container used in the soft food-contained package.

[0008] FIG. 3 is an explanatory view schematically illustrating the packaging containers in a stacked state.

[0009] FIG. 4 is an explanatory view illustrating a structure of the packaging container.

[0010] FIG. 5 is a cross-sectional view schematically illustrating a structure of the soft food-contained package.

[0011] FIG. 6 is a cross-sectional view schematically illustrating a structure of another example of the soft food-contained package.

[0012] FIG. 7 is a cross-sectional view schematically illustrating a structure of another example of the soft food-contained package.

[0013] FIG. 8 is a cross-sectional view schematically illustrating a structure of another example of the soft food-contained package.

[0014] FIG. 9 is a cross-sectional view schematically illustrating a structure of another example of

the soft food-contained package.

[0015] FIG. **10** is a cross-sectional view schematically illustrating a structure of another example of the soft food-contained package.

[0016] FIG. **11** is a plan view illustrating a structure of another example of the packaging container.

DETAILED DESCRIPTION

[0017] Hereinafter, a structure of a soft food-contained packaging container **11** according to an embodiment of the present invention, and a structure of the packaging container **11** used in a package **1** are described with reference to FIG. **1** to FIG. **10**.

[0018] FIG. **1** is a side view illustrating a structure of the soft food-contained package **1** according to an embodiment of the present invention, and illustrating, by a cross section of a part of the packaging container **11** of the soft food-contained package **1**, part of a food **13** contained therein. FIG. **2** is a plan view illustrating a structure of the packaging container **11**. FIG. **3** is an explanatory view schematically illustrating two packaging containers **11** in a stacked state. FIG. **4** is an explanatory view illustrating a structure of the packaging container **11**. FIG. **5** is a cross-sectional view schematically illustrating a structure of the soft food-contained package **1**, and FIG. **6** to FIG. **10** are cross-sectional views schematically illustrating structures of other examples of the soft food-contained package **1**.

[0019] As illustrated in FIG. **1**, the soft food-contained package **1** includes the packaging container **11**, a cover **12**, and the food **13** including a soft food **31** and a liquid **32**. The soft food-contained package **1** is a product in which the soft food **31** and liquid **32** are filled in the packaging container **11**, and the packaging container **11** is sealed by the cover **12**. The packaging container **11** and the cover **12** form a package that contains the food **13**. In addition, the soft food-contained package **1** packages the food **13** (soft food **31** and liquid **32**) by the cover **12** sealing the packaging container **11** in a state in which the soft food **31** and liquid **32** are filled in the packaging container **11**.

[0020] The soft food-contained package **1** is formed to be capable of cooking with heat, for example, by a microwave oven and/or by warming in hot water. In addition, the soft food-contained package **1** is formed to be capable of, for example, storage at room temperature, storage by refrigeration, and/or storage by freezing. Note that in the present embodiment, the description is given by defining the up-and-down direction such that the packaging container **11** side is a downward side, and the cover **12** side is an upward side.

[0021] As illustrated in FIG. **1** to FIG. **3**, the packaging container **11** includes a receptacle portion **21**, a flange portion **22**, a ridge seal portion **23**, and a liquid-splash prevention projecting portion **24**. The packaging container **11** is formed by shaping a multilayer thermoplastic resin sheet. Specifically, in the packaging container **11**, the receptacle portion **21**, flange portion **22**, ridge seal portion **23** and liquid-splash prevention projecting portion **24** are formed of the thermoplastic resin sheet as one piece. In addition, a typical forming method for forming the thermoplastic resin sheet into the packaging container **11** is air pressure forming or injection molding, but other methods may be adopted. The packaging container **11** has, for example, a self-standing property by the receptacle portion **21**. Note that the packaging container **11** may be configured to have no self-standing property.

[0022] In addition, the thermoplastic resin sheet that forms the packaging container **11** may be formed of a single-layer resin sheet or a multilayer resin sheet if the ridge seal portion **23**, which is a part to which the cover **12** is heat-sealed, is formed of a resin with heat-seal property. It is preferable that the packaging container **11** is formed of a multilayer structure including, as an outer layer, a resin layer with heat seal property. In addition, it is preferable that the packaging container **11** has a gas barrier property, in addition to the heat seal property.

[0023] In the present embodiment, the packaging container **11** is formed of a multilayer thermoplastic resin sheet in which inner and outer surface layers of polypropylene and an intermediate layer of an ethylene vinyl alcohol copolymer are bonded via an adhesive layer. Note that the surface layer is formed by using, for example, polyethylene, polypropylene, ethylene-

propylene copolymer, or an olefin-based resin with a heat seal property, such as a blend of these. In addition, the intermediate layer is formed by using, for example, a thermoplastic resin with a gas barrier property such as Mx nylon, vinylidene chloride-based resin, or nylon-based resin. The surface layer and the intermediate layer are bonded via an adhesive layer formed of a resin having thermal adhesivity to both, such as acid modified olefin-based resin, copolyester-based adhesive resin, or epoxy modified thermoplastic adhesive resin.

[0024] In the receptacle portion **21**, a plurality of recesses that are filled with the soft food **31** are formed. In a concrete example, the receptacle portion **21** includes a side wall portion **25**, a planar portion **26** and a recess portion **27**.

[0025] The side wall portion **25** is integrally continuous with the flange portion **22** at an open end (upper end), and is integrally continuous with the planar portion **26** at a lower end. In the present embodiment, the side wall portion **25** is formed, for example, in a cylindrical shape. In a concrete example, the side wall portion **25** is formed in such a cylindrical shape as to have an identical diameter from the upper end to a central side in the up-and-down direction, to have a reduced diameter at the central side, and to have a gradually reduced diameter from the central side toward the lower end.

[0026] The planar portion **26** is formed in a planar shape extending in a direction perpendicular to the axial direction of the side wall portion **25**. The planar portion **26** is formed to have a circular outer peripheral edge. The recess portion **27** is formed by the planar portion **26** being recessed at the location.

[0027] For example, a plurality of recess portions **27** are provided. It is preferable that three or more recess portions **27**, for example, are provided. For example, the recess portions **27** are formed in different shapes. Here, the different shapes relate to shapes of openings of the recess portions **27**, shapes of side surfaces and bottom surfaces of the recess portions **27**, the depths of the recess portions **27**, and the like. Note that all recess portions **27** may not have different shapes, and it suffices that the recess portions **27** have at least two or more kinds of shapes.

[0028] In the example of the present embodiment, three recess portions **27** are provided, and the three recess portions **27** have an equal depth, and have different shapes except for the depth. In addition, in this example, the side walls of the three recess portions **27** are inclined such that the opening shape becomes smaller from the upper-end opening toward the bottom portion. Note that the side wall may have a shape along the up-and-down direction.

[0029] For example, one of the three recess portions **27** is a pocket that is recessed in an omelet shape, another is a pocket that is recessed in a wienerwurst shape or an arcuate shape, and the other is a pocket that is recessed in a substantially triangular shape with curved corners. Note that the shapes of the recess portions **27** are not limited. For example, like a recess portion **27** of another embodiment illustrated in FIG. **11**, the pocket may be recessed not in the omelet shape, but in a shape simulating a slice of fish.

[0030] The recess portions **27** are arranged on the planar portion **26** in such a manner that a part of the planar portion **26** is present between mutually neighboring recess portions **27**. In other words, the recess portions **27** are arranged on the planar portion **26** by being spaced apart.

[0031] In addition, the recess portions **27** are arranged on the planar portion **26** such that at least one of the recess portions and a part of the planar portion **26** are present in each of two orthogonal directions along the plane direction of the planar portion **26**. In other words, the recess portions **27** are arranged such that at least one of the recess portions **27** is disposed in a radiating direction (radial direction) from the center of the planar portion **26** and a direction along the radiating direction (radial direction).

[0032] Specifically, like a plurality of recess portions **271** of a comparative example indicated by two-dot-and-dash lines in FIG. **4**, if the recess portions **271** are uniformly arranged, there occurs an area A in one of directions, where only a planar portion **26** is present and the recess portion **271** is not disposed. By contrast, at least one of the recess portions **27** of the embodiment is disposed in

the radiating direction (radial direction) from the center of the planar portion **26** and the direction along the radiating direction (radial direction), and thereby the recess portions **27** are formed on the planar portion **26** with such a non-uniform arrangement that the area A where the recess portion **27** is not disposed does not occur on the planar portion **26**, except for the outer peripheral edge of the planar portion **26**.

[0033] In addition, in the case where the packaging container **11** is configured to have a self-standing property, the recess portions **27** may acquire a self-standing property by a bottom portion of any one of the recess portions **27**, or may acquire a self-standing property by bottom portions of two or more of the recess portions **27**.

[0034] The flange portion **22** is provided on the receptacle portion **21**. The flange portion **22** extends outward from an opening end of the side wall portion **25** of the receptacle portion **21**. The shape of the outer peripheral edge of the flange portion **22** is set to be a rectangular shape, a circular shape, an oval shape, or a modified shape such as a combination of corner portions and a circular shape. In the present embodiment, the flange portion **22** forms a square outer edge shape with four arcuate corner portions, and the shape of the inner peripheral edge that is continuous with the opening end of the receptacle portion **21** is circular.

[0035] In addition, in the flange portion **22**, the ridge seal portion **23** and the liquid-splash prevention portion **24**, which project from an upper surface of the flange portion **22** to an upward side that is an opposite side to the receptacle portion **21** side (the side of the planar portion **26** and recess portions **27**), are formed. Specifically, the flange portion **22** includes the ridge seal portion **23** and the liquid-splash prevention portion **24**, which are integrally formed at a time of forming the packaging container **11**.

[0036] The ridge seal portion **23** projects upward from a major surface of the flange portion **22**. The ridge seal portion **23** is formed in an annular shape that is continuous with the upper surface of the flange portion **22**. In other words, the ridge seal portion **23** extends over the entire circumference of the flange portion **22** in such a manner as to surround the inner peripheral edge of the flange portion **22** (the opening end of the receptacle portion **21**).

[0037] An upper surface of the ridge seal portion **23** is formed in a planar shape. The upper surface of the ridge seal portion **23** is a seal surface on which the cover **12** is heat-sealed and with which the cover **12** is put in close contact. The ridge seal portion **23** projects upward from the major surface of the flange portion **22** with a predetermined width and height. Note that the width and height of the ridge seal portion **23** are properly set based on various conditions such as formability, sealability and openability of the cover **12**, and the like. Specifically, the seal surface of the ridge seal portion **23** is set to have a predetermined seal area, and good sealability and openability can be secured. In addition, here, the width of the ridge seal portion **23** means a dimension in a direction that extends along the major surface of the flange portion **22** and is perpendicular to the extending direction of the ridge seal portion **23**.

[0038] In addition, the ridge seal portion **23** is provided with a beak portion (first projecting portion) **23a** at a position corresponding to an opening start portion at which the cover **12** is opened. As illustrated in FIG. 2, in the present embodiment, beak portions (first projecting portions) **23a** are provided at four locations corresponding to the corner portions of the square-shaped flange portion **22** having curved corner portions. Note that the beak portion **23a** is provided at the opening start portion of the cover **12** of the ridge seal portion **23**, and, for example, if the opening start portion is provided at one location, it suffices that the beak portion **23a** is provided at one location of the ridge seal portion **23**. Specifically, the opening start portion is appropriately set in consideration of the shape, the purpose of use, or the like of the packaging container **11**, and the beak portion **23a** is provided at a part of the ridge seal portion **23** in accordance with the opening start portion that is set.

[0039] The beak portion **23a** is formed in a ridge shape projecting outward from the flange portion **22**. Specifically, the beak portion **23a** is a projection portion that is formed such that a part of the

ridge seal portion **23** projects outward in the radial direction of the opening portion of the receptacle portion **21**. In a concrete example, the beak portion **23a** is formed to have a width and height similar to the width and height of the ridge seal portion **23**, and is formed in a shape with a pointed tip. Here, the shape with the pointed tip of the beak portion **23a** is a beak shape, an angular shape, an angle bracket (<) shape (V shape), or the like.

[0040] The beak portion **23a** may not have an angular shape or a beak shape with a pointed tip, if stress concentrates at the distal end of the beak portion and the cover is easily pulled off with the beak portion **23a** being the opening start point, when the cover **12** that is heat-sealed to the ridge seal portion **23** is pulled off from the packaging container **11**. Specifically, the beak portion **23a** can be appropriately set to have a curved shape without a pointed distal end, or a shape with a projecting distal end, if the force for opening the cover **12** can be concentrated by reducing the contact area that is in contact with the cover **12**, when the cover **12** is moved in such a direction as to peel off the cover **12** at the opening start point.

[0041] The liquid-splash prevention projecting portion (second projecting portion) **24** is provided at a predetermined distance from the beak portion **23a**, on the inner peripheral side of the beak portion **23a**. The liquid-splash prevention projecting portion **24** includes a side surface extending along the inner surface (inside surface) of the beak portion **23a**, and is disposed on the flange portion **22** by being spaced apart from the beak portion **23a** with a predetermined gap. For example, the gap between the beak portion **23a** and the liquid-splash prevention projecting portion **24** is set at about 1 mm.

[0042] By the liquid-splash prevention projecting portion **24** being disposed with the predetermined gap from the beak portion **23a** of the ridge seal portion **23**, a gap is formed along the inner surface of the beak portion **23a** between the liquid-splash prevention projecting portion **24** and the inner peripheral side of the beak portion **23a**.

[0043] For example, an upper surface of the liquid-splash prevention projecting portion **24** is formed in a planar shape. The upper surface of the liquid-splash prevention projecting portion **24** serves as a seal surface, and the cover **12** is heat-sealed on the upper surface when the cover **12** and the ridge seal portion **23** are heat-sealed.

[0044] Both ends of the liquid-splash prevention projecting portion **24** may be formed to be spaced apart from the beak portion **23a** (ridge seal portion **23**), or one end thereof may be spaced apart from the beak portion **23a** (ridge seal portion **23**) and the other end thereof may be formed to be continuous with the beak portion **23a** (ridge seal portion **23**).

[0045] For example, as illustrated in FIG. 2, the liquid-splash prevention projecting portion **24** is formed in an angle bracket (<) (V shape) along the beak portion **23a**. In the case of adopting this liquid-splash prevention projecting portion **24**, the opening start portion of the flange portion **22** is configured to be provided with a double beak portion by the beak portion **23a** and the liquid-splash prevention projecting portion **24**. Note that the liquid-splash prevention projecting portion **24** may have a triangular shape.

[0046] Note that in the case where the inside surface of the ridge seal portion **23** and the side surface of the liquid-splash prevention projecting portion **24** extend in an inclined direction to the major surface of the flange portion **22**, it suffices that the extending direction of the side surface of the liquid-splash prevention projecting portion **24** extends along the extending direction of the beak portion **23a**, and the directions of inclination of the inside surface of the beak portion **23a** and the side surface of the liquid-splash prevention projecting portion **24** may be different.

[0047] In other words, it suffices that a ridgeline (ridge portion) of the liquid-splash prevention projecting portion **24** on the beak portion **23a** side has a shape along the extending direction of the beak portion **23a**.

[0048] The height of the liquid-splash prevention projecting portion **24** is set to be equal to or less than the height of the ridge seal portion **23** (beak portion **23a**). Preferably, the height of the liquid-splash prevention projecting portion **24** is set to be less than the height of the ridge seal portion **23**

(beak portion **23a**). Note that the height of the liquid-splash prevention projecting portion **24** is preferably set to be such a height as to be capable of suppressing a decrease in liquid-splash prevention effect, increasing a heat-seal strength between the liquid-splash prevention projecting portion **24** and the cover **12**, and suppressing a decrease in openability. Specifically, the liquid-splash prevention projecting portion **24** is preferably set to have such a height that the liquid-splash prevention effect is obtained, the adhesive force between the liquid-splash prevention projecting portion **24** and the cover **12** is weak, and the cover **12** can smoothly be peeled off with no resistance at a time of peeling the cover **12**.

[0049] The width of the liquid-splash prevention projecting portion **24** is appropriately set. The width of the liquid-splash prevention projecting portion **24** is appropriately set, for example, based on the adhesive force in the case where the cover **12** and the liquid-splash prevention projecting portion **24** are adhered by heat-seal. Specifically, the width of the liquid-splash prevention projecting portion **24** is set based on the height of the liquid-splash prevention projecting portion **24**, the area and shape of the upper surface, and the like. Note that, here, the width of the liquid-splash prevention projecting portion **24** means a dimension in a direction that extends along the major surface of the flange portion **22** and is perpendicular to the extending direction of the liquid-splash prevention projecting portion **24**.

[0050] Note that the inside surface (inner surface) of the ridge seal portion **23** and that side surface of the liquid-splash prevention projecting portion **24**, which is opposed to the ridge seal portion **23** (beak portion **23a**) may extend in a direction perpendicular to the major surface of the flange portion **22**, or may extend in a direction inclined to the major surface of the flange portion **22**.

[0051] The cover **12** seals the receptacle portion **21** by being fixed to the flange portion **22** and covering the receptacle portion **21**. The cover **12** is melt-adhered to the ridge seal portion and liquid-splash prevention projection portion **24**, which are formed on the flange portion **22** by heat-seal in the packaging container **11**. The cover **12** is formed in a film shape. In one example, the cover **12** is formed of a multilayer film of PET/adhesive layer (modified polypropylene)/cast polypropylene.

[0052] Note that the structure of the cover **12** is not limited to the structure in which the cover **12** is melt-adhered to the ridge seal portion **23** and the liquid-splash prevention projecting portion **24**. For example, the ridge seal portion **23** and the liquid-splash prevention projecting portion **24** may not be formed on the flange portion **22**, and the cover **12** may be configured to be directly melt-adhered to the flange portion **22**. In addition, the cover **12** may not be melt-adhered, and the cover **12** may be formed in a cup shape or a dish shape and may be engaged with an outer peripheral edge of the flange portion **22**. Besides, the cover **12** may be configured to be engaged with an inner peripheral edge of the flange portion **22**, that is, an inner edge of an upper end of the side wall portion **25** of the receptacle portion **21**.

[0053] The food **13** includes at least the soft food **31** and the liquid **32**. The food **13** is, for example, a universal design food that is provided as a food for a person with difficulty in swallowing or as a food for a person with difficulty in chewing. Universal design foods are classified into four classifications in accordance with hardness or the like, namely “easily chewable”, “crushable by the gums”, “crushable by the tongue”, and “no need of chewing”. The food of the classification “easily chewable” requires a hardness of 5×10^5 N/m² or less. The food of the classification “crushable by the gums” requires a hardness of 5×10^4 N/m² or less. The food of the classification “crushable by the tongue” requires a hardness of 2×10^4 N/m² or less. The food of the classification “no need of chewing” requires a hardness of 5×10^3 N/m² or less.

[0054] The soft food **31** is, for example, a mousse-like food. The soft food **31** may be, like a custard pudding, a food obtained by solidifying a liquid including protein by utilizing thermal denaturation of protein.

[0055] The mousse-like food is a food obtained by kneading and crushing a solid food, mixing the

resultant with a coagulant or the like, and then forming the mixture into a shape of a part in which the recess portion **27** is accommodated.

[0056] The solid food is, for example, an animal food material, a vegetable food material, or a combination thereof. The solid food can include meat, fish and shellfish, seaweed, fruit, nuts, cereal, vegetables, mushrooms, or two or more of these. The solid food may be a food material before cooking, or a cooked food. In the case of the latter, the solid food may further include one or more of oils and fats, and seasonings such as sugar and salt.

[0057] Preferably, the solid food is mashed to a desired grain size. As the grain size becomes smaller, the mouth-feel of the soft food **31** becomes smoother. The grain size of the mashed food is preferably 2.0 mm or less, more preferably 0.5 mm or less, and still more preferably 0.1 mm or less. Here, the “grain size” is a value obtained by grain size distribution measurement by a laser diffraction-scattering method.

[0058] The coagulant is, for example, a protein that is thermally denatured by heat and irreversibly solidified. Such a protein is, for example, mixed with the mashed food in a form of an aqueous solution. As the protein or aqueous solution thereof, for example, the white of an egg can be used. The protein is added such that the concentration in the soft food **31** is preferably within a range of 0.1 to 25 mass %, and more preferably within a range of 1 to 15 mass %.

[0059] The mixture of the mashed food, coagulant and the like is, for example, formed into a shape similar to the food before mashing or a food whose taste is to be reproduced by the soft food **31**. For example, the shape of the recess portion **27** of the packaging container **11** is set to be a shape similar to the food whose taste is to be reproduced, and the mixture is filled in the recess portion **27** and is heated. The heating is performed preferably at temperatures within a range of 50 to 130° C., more preferably at temperatures within a range of 60 to 130° C., and still more preferably at temperatures within a range of 80 to 130° C. Thereby, thermal denaturing of the protein is caused, and the mixture is coagulated. In addition, the heating of the mixture filled in the recess portion **27** may be performed before the sealing of the packaging container **11** by the cover **12**, or may be performed after the sealing of the packaging container **11** by the cover **12**. Note that the heating of the mixture may be performed before the filling in the recess portion **27**.

[0060] Preferably, the soft food **31** has a hardness of $5 \times 10^{2.4}$ N/m² or less. The hardness of the soft food **31** may be $5 \times 10^{2.3}$ N/m² or less. The problem of the loss of shape becomes conspicuous in the case where the softness of the soft food **31** is high. The method described here is particularly useful in the case where the hardness of the soft food **31** is low.

[0061] Preferably, the soft food **31** has a hardness of $2 \times 10^{2.3}$ N/m² or more. If the soft food **31** is too soft, it becomes difficult for the soft food **31** to keep the shape. The soft food **31** may have a hardness of $1 \times 10^{2.4}$ N/m² or more.

[0062] The soft food **31** is formed up to a predetermined height from the bottom surface of the recess portion **27**. For example, the soft food **31** may be formed up to an upper end of the recess portion **27**, may be formed up to below the upper end of the recess portion **27**, or may be formed up to above the recess portion **27**, i.e., above the planar portion **26**, and below the upper end of the side wall portion **25**. In other words, the height of the soft food **31** is lower than the height from the bottom portion of the soft food **31** (the bottom surface of the recess portion **27**) to the planar portion **26**, or is equal to the height from the bottom portion of the soft food **31** (the bottom surface of the recess portion **27**) to the planar portion **26**, or is higher than the height from the bottom portion of the soft food **31** (the bottom surface of the recess portion **27**) to the planar portion **26** and is lower than the height up to the upper end of the side wall portion **25** (the lower surface of the cover **12**). In addition, the soft food **31** is formed in close contact with the recess portion **27** without a gap. In other words, the soft food **31** is formed in close contact with the bottom surface and side surface of the recess portion **27**.

[0063] A description is given of the relationship between the soft food **31**, and a head space occurring above the soft food **31** in the inside of the packaging container **11** and cover **12** that are

the package. As illustrated in FIG. 5, the height (the height of the soft food **31**) from the lower surface of the soft food **31** formed in the recess portion **27** (the bottom surface of the recess portion **27**) to the upper surface of the soft food **31** is defined as H1. In addition, the height (the height of the head space) from the upper surface of the soft food **31** to the lower surface of the cover **12** is defined as H2. At this time, the height H1 of the soft food **31** and the height H2 of the head space have a relationship of $H1 > H2$.

[0064] The liquid **32** is water, a seasoning liquid, rice porridge, or the like. In the example of the present embodiment, the liquid **32** is a seasoning liquid, and the description below is given on the assumption that the liquid **32** is a seasoning liquid **32**. The seasoning liquid **32** is a liquid including water, and a seasoning, an extract of an animal food material, an extract of vegetable food material, fat, or a combination thereof. The seasoning liquid **32** is, for example, a solution, a dispersion liquid such as a sol, or a combination thereof.

[0065] The seasoning liquid **32** can further include a thickener. Examples of the thickener include, although not limited to these, guar gum, xanthan gum, tamarind gum, carrageenan, carboxymethyl cellulose, Tara gum, locust bean gum, curdlan, sodium alginate, psyllium seed gum, gellan gum, gum arabic, karaya gum, chitin, chitosan, arabinogalactan, welan gum, Sesbania gum, aloe vera extract, dammar resin, ghatti gum, elemi resin, dextran, oligoglucosamine, Abelmoschus Manihot, microfibrinous cellulose, tragacanth gum, natto bacteria gum, Fukuronori extract, Furcellaria, Cassia gum, Macrophomopsis gum, pullulan, aloe arborescens extract, peach resin, Aureobasidium culture solution, guar gum enzymatic decomposition product, rhamsan gum, glucosamine, Levan, agrobacterium succinoglycan, yeast cell membrane, linseed gum, artemisia seed gum, almond gum, sclerogum, or two or more of these.

[0066] Preferably, the thickener is guar gum, xanthan gum, tamarind gum, carboxymethyl cellulose, Tara gum, locust bean gum, sodium alginate, psyllium seed gum, gum arabic, karaya gum, welan gum, tragacanth gum, natto bacteria gum, Furcellaria, Cassia gum, peach resin, guar gum enzymatic decomposition product, rhamsan gum, agrobacterium succinoglycan, or two or more of these.

[0067] More preferably, the thickener is guar gum, xanthan gum, tamarind gum, carboxymethyl cellulose, Tara gum, locust bean gum, sodium alginate, psyllium seed gum, welan gum, tragacanth gum, rhamsan gum, agrobacterium succinoglycan, or two or more of these.

[0068] According to one example, the seasoning liquid **32** includes, as the thickener, xanthan gum and guar gum. In addition, according to one example, the concentration of the thickener in the food **13** is in a range of 0.01 mass % to 3.00 mass %.

[0069] The seasoning liquid **32** may further include a gelling agent. The gelling agent is, for example, pectin, gelatine, agar, or two or more of these. The concentration of the gelling agent in the seasoning liquid **32** is preferably 1.0 mass % or less, more preferably 0.2 mass % or less, and still more preferably 0.1 mass % or less.

[0070] The viscosity of the seasoning liquid **32** is preferably within a range of 0.3 to 20.0 Pa.Math.s, more preferably within a range of 0.3 to 10.0 Pa.Math.s, and still more preferably within a range of 1.5 to 2.0 Pa.Math.s. The solution or sol in the above-described classifications “crushable by the tongue” and “no need of chewing” requires 1500 mPa.Math.s or more. Note that the sol of the classification “crushable by the tongue” requires a hardness of 1×10^4 N/m² or less. Besides, the sol of the classification “no need of chewing” requires a hardness of 3×10^3 N/m² or less.

[0071] As the viscosity of the seasoning liquid **32** becomes higher, the vibration of the surface of the soft food **31** and the vibration of the seasoning liquid **32** are more reduced at the time of, for example, transportation of the soft food-contained package **1**. In addition, as the viscosity of the seasoning liquid **32** becomes higher, the movement of the soft food **31** in the packaging container **11** is more suppressed, and the contact of the soft food **31** with the inner surface of the container less easily occurs. Accordingly, the movement of the soft food **31** from the recess portion **27** into

the head space between the upper surface of the soft food **31** and the lower surface of the cover **12**, and the loss of shape of the soft food **31**, which occur due to these factors, will occur less easily. [0072] Note that the viscosity of the seasoning liquid **32** is calculated by using a B-type rotational viscometer, setting a measurement temperature at $20\pm 2^{\circ}\text{C.}$, rotating an M3 rotor or M4 rotor at 12 rpm, reading an indication after two minutes, multiplying the value of the indication by a corresponding coefficient, and expressing the obtained value by Pa.Math.s.

[0073] The seasoning liquid **32** is filled in the head space occurring above the soft food **31** in the packaging container **11**. The seasoning liquid **32** is filled in the packaging container **11** in such an amount that the liquid surface is set at a position equal to or higher than the planar portion **26**. The seasoning liquid **32** may be filled in an amount less than the capacity of the head space, or may be filled in an equal amount to the capacity of the head space. Specifically, the seasoning liquid **32** may be filled up to a height below the upper end of the side wall portion **25** (the lower surface of the cover **12**) in such a manner that a space occurs in an upper part of the head space. Alternatively, the seasoning liquid **32** may be filled up to the same position as the upper end of the side wall portion **25** (the lower surface of the cover **12**), and may fill the head space such that no gas is present in the head space.

[0074] The food **13** as described above is, for example, a main dish and/or a side dish. Examples of main dishes and side dishes, which are reproduced by the soft food **31** and seasoning liquid **32** of the food **13**, are, although not limited to these, curry, stew, boiled vegetables, and boiled food such as sukiyaki, or a hamburger, tomato chicken stew, stir-fried meat with vegetables, a wienerwurst, fish boiled with sugar and soy sauce, grilled fish, an omelet, and combinations of a grilled food, a stir-fried food or steamed food of vegetables, tofu or the like, and sauces.

[0075] Next, a plurality of examples of the soft food **31** and seasoning liquid **32**, which are filled in the packaging container **11** of the soft food-contained package **1**, are described with reference to FIG. 5 to FIG. 10. Note that the soft food **31** and seasoning liquid **32** are not limited to the examples to be described below. In addition, FIG. 5 to FIG. 10 schematically illustrate the packaging container **11**, with partial structures being omitted.

[0076] FIG. 5 illustrates a soft food-contained package **1** of a first example. As illustrated in FIG. 5, as the soft food **31** contained in the recess portions **27**, different kinds of soft foods **31A** and **31B** are contained. Here, the different kinds of soft foods are two or more kinds of soft foods with different ingredients, and the different ingredients mean that at least a part of ingredients is different. Specifically, the different kinds of soft foods mean that, of two or more soft foods, at least one soft food includes, among ingredients, an ingredient different from ingredients of another soft food, or that all ingredients of one soft food are formed of ingredients different from the ingredients of another soft food. In addition, the meaning of the different ingredient includes that the ingredient itself is different, or that the amount of the ingredient is different. Besides, each of the soft foods **31A** and **31B** is filled up to a height of an intermediate part of each recess portion **27** in the height direction. The seasoning liquid **32** is filled in the head space in the package, and is filled up to a position lower than the upper end of the side wall portion **25** (the lower surface of the cover **12**).

[0077] FIG. 6 illustrates a soft food-contained package **1** of a second example. As illustrated in FIG. 6, as the soft food **31** contained in the recess portions **27**, the same kind of soft food **31** is contained. The soft food **31** is filled up to above the planar portion **26** beyond each recess portion **27**. The seasoning liquid **32** is filled in the head space in the package, and is filled up to a position lower than the upper end of the side wall portion **25** (the lower surface of the cover **12**).

[0078] FIG. 7 illustrates a soft food-contained package **1** of a third example. As illustrated in FIG. 7, as the soft food **31** contained in the recess portions **27**, different kinds of soft foods **31A** and **31B** are contained. In a concrete example, as illustrated in FIG. 7, a first soft food **31A** is filled as the soft food **31** contained in one recess portion **27**, a second soft food **31B** is filled in a lower layer as the soft food **31** contained in the other recess portion **27**, and the first soft food **31A** is filled in an upper layer as the soft food **31** contained in the other recess portion **27**. In addition, the first soft

food **31A** is filled up to above the planar portion **26** beyond each recess portion **27**. The seasoning liquid **32** is filled in the head space in the package, and is filled up to a position lower than the upper end of the side wall portion **25** (the lower surface of the cover **12**).

[0079] FIG. **8** illustrates a soft food-contained package **1** of a fourth example. As illustrated in FIG. **8**, the soft food **31** is contained in one or more recess portions **27** among the recess portions **27**, and the soft food **31** is not filled in at least one recess portion **27**. The soft food **31** contained in recess portions may be the same food **31** or different foods. In a concrete example, as illustrated in FIG. **8**, the soft food **31** is filled in at least one recess portion **27** up to a height of an intermediate part of the recess portion **27** in the height direction. The seasoning liquid **32** is filled up to upper ends of the recess portion **27** filled with the soft food **31** and the recess portion **27** not filled with the soft food **31**, that is, up to the planar portion **26**.

[0080] FIG. **9** illustrates a soft food-contained package **1** of a fifth example. As illustrated in FIG. **9**, as the soft food **31** contained in the recess portions **27**, different kinds of soft foods **31A** and **31B** are contained. In addition, the first soft food **31A** is filled up to a height of an intermediate part of the recess portion **27** in the height direction, and the second soft food **31B** is filled up to the upper end of the recess portion **27** (the planar portion **26**). The seasoning liquid **32** is filled in the head space in the package, and is filled up to a position lower than the upper end of the side wall portion **25** (the lower surface of the cover **12**). In addition, the soft food **31B** includes content **31a** having a different taste, color and/or hardness from a mousse-like food that forms the soft food **31B**. The content **31a** is, for example, a mousse-like food that forms the soft food **31B**. Note that the content **31a** is not limited to the mousse-like food, and may be various foods, such as a solid food, if a target person who eats the soft food **31B** can eat. Besides, for example, the content **31a** may be a solidified seasoning liquid **32** that has a different taste from the seasoning liquid **32** and is solidified by a gelling agent.

[0081] FIG. **10** illustrates a soft food-contained package **1** of a sixth example. As illustrated in FIG. **10**, as the soft food **31** contained in a plurality of recess portions **27**, the same kind or different kinds of soft foods **31A** are contained. For example, the soft food **31A** is filled up to above the planar portion **26** beyond each recess portion **27**. A seasoning liquid **32A** is filled in the head space in the package, and is filled up to a position lower than the upper end of the side wall portion **25** (the lower surface of the cover **12**). In addition, the packaging container **11** includes one or more recess portions **28** provided in the flange portion **22**, in addition to the recess portions **27** formed on the planar portion **26**, and a soft food **31B** and/or a seasoning liquid **32B** is contained in the recess portion **28**. For example, the soft food **31A** and seasoning liquid **32A** contained in the recess portions **27** are a main dish and/or a side dish, and the soft food **31B** and seasoning liquid **32B** contained in the recess portions **28** provided in the flange portion **22** are a main dish and/or a side dish of a different taste, or sweets such as a custard pudding, a jerry, or a mousse cake.

[0082] As illustrated in these examples, in regard to the soft food **31** and seasoning liquid **32** contained in the soft food-contained package **1**, if the height H1 of the soft food **31** is higher than the height of the head space H2 and the seasoning liquid **32** is filled up to at least the planar portion **26**, the soft food-contained package **1** can have various structures. In addition, the above-described examples are merely examples, and the structures of the various examples can be combined. Specifically, the seasoning liquid **32** in the example 1 and the example 2 can be filled until filling the head space. In addition, two or more kinds of soft foods **31** may be filled in one recess portion **27**, and may be stacked in two or more layers, and one or more kinds of contents **31a** may be included in the soft food **31**. Furthermore, the content **31a** may be configured to be included in the seasoning liquid **32**. Besides, two or more different kinds of seasoning liquids **32** may be filled, and may be formed in a mixed state in a common head space occurring above the recess portions **27**.

[0083] According to the soft food-contained package **1** with the above-described structure, the soft food **31** is contained in the recess **27** in the shape of the recess portion **27**, and the seasoning liquid **32** is contained in the head space occurring above the soft food **31**. Thereby, the area in which the

soft food **31** is in contact with the seasoning liquid **32** is the upper surface of the soft food **31**. Specifically, in the soft food-contained package **1**, since the soft food **31** is filled in the recess portion **27**, the area of contact of the soft food **31** with the seasoning liquid **32** can be reduced. Thus, the taste and/or color of the seasoning liquid **32** can be prevented from excessively spreading to the soft food **31**. Note that in a case where the taste and/or color of the seasoning liquid **32** is lighter than the taste and/or color of the soft food **31**, the taste and/or color of the soft food **31** can be prevented from excessively spreading to the seasoning liquid **32**.

[0084] Additionally, in the soft food-contained package **1**, if the height of the soft food **31** filled in the recess portion **27** is set to be equal to or less than the height of the planar portion **26**, the area of contact of the seasoning liquid **32** with each soft food **31** can be decreased as much as possible, thus being able to suppress the spreading of the taste of the soft food **31** into the seasoning liquid **32**, and the spreading of the taste and color of the seasoning liquid **32** into the soft food **31**.

[0085] Additionally, the soft food **31** is formed in the shape of the recess portion **27**, and is contained in the recess portion **27**. Thus, even if vibration or impact is applied to the soft food-contained package **1** at a time of transportation, the soft food **31** does not easily move and is prevented from being crumbled. In addition, since the soft food **31** is contained in the recess portion **27**, drips occurring from the soft food **31** after the formation of the soft food **31** mix with the seasoning liquid **32**, and thus the drips occurring from the soft food **31** are not visually recognized.

[0086] Additionally, even in the case where different soft foods **31** are contained in the recess portions **27**, the seasoning liquid **32** is in contact with each soft food **31**. Thereby, even in the case where different soft foods **31** are treated, the soft food-contained package **1** is configured such that the seasoning liquid **32** is in contact with the upper surface of each soft food **31**, and thus the flavor or taste spreading from each soft food **31** decreases. Thus, the soft food-contained package **1** can achieve proper harmony of taste of the seasoning liquid **32**, and can suppress excessive mixing of state. Furthermore, because of the structure in which the seasoning liquid **32** is in contact with only the upper surface of the soft food **31**, it is possible to reduce excessive permeation or a change in taste or hardness of the soft food **31** by the seasoning liquid **32**.

[0087] Additionally, the soft food-contained package **1** is configured such that the height H1 of the soft food **31** is higher than the height H2 of the head space. At a time of transporting the soft food-contained package **1**, for example, when the soft food-contained package **1** is vertically inverted, that is, when the receptacle portion **21** is turned to the upper side and the cover **12** is turned to the lower side, even if the soft food **31** is separated from the recess portion **27**, the soft food **31** does not completely slip out of the recess portion **27** and a part of the soft food **31** stays in the recess portion **27**. Thereby, when the attitude of the soft food-contained package **1** is restored, that is, when the receptacle portion **21** is turned back to the lower side and the cover **12** is turned back to the upper side, the soft food **31** is contained in the recess portion **27** once again.

[0088] Additionally, the liquid surface of the seasoning liquid **32** is set at the same position as the upper surface of the planar portion **26** or is set above the planar portion **26**. Thereby, even if the soft food-contained package **1** is inclined and the seasoning liquid **32** moves, if the attitude is restored, the seasoning liquid **32** can come in contact with the upper surface of the soft food **31** provided in each recess portion **27**.

[0089] Additionally, the packaging container **11** includes the planar portion **26**, and a part of the planar portion **26** is disposed between the mutually neighboring recess portions **27**. Thus, compared to the structure in which the recess portions **27** are continuous, the strength can be increased.

[0090] Additionally, in the packaging container **11**, the recess portions **27** provided on the planar portion **26** are arranged such that at least one of the recess portions **27** and the planar portion **26** are disposed in each of two orthogonal directions along the plane direction of the planar portion **26**. Thereby, the strength of the packaging container **11** can be increased. Specifically, as indicated by two-dot-and dash lines in FIG. 4, in the structure in which the recess portions **271** are arranged by

being uniformly spaced apart, the area A where the recess portion 271 is not disposed occurs in one direction. Then, since the strength of this area A is the strength of the planar portion 26, if external force is applied and exceeds the strength of the planar portion 26, there occurs bending deformation in the area A. By contrast, as illustrated in FIG. 1, except for the outer peripheral edge of the planar portion 26, there occurs no such area A where only the planar portion 26 is disposed and the recess portion 27 is not disposed in the direction along the plane direction of the planar portion 26. Thereby, the packaging container 11 can improve the strength of the planar portion 26 by the recess portions 27.

[0091] Additionally, although the number of recess portions 27 provided in the packaging container 11 is not limited, by providing three or more recess portions 27, it becomes easier to improve the strength of the planar portion 26, and to arrange at least one recess portion in the plane direction of the planar portion 26. Furthermore, in the soft food-contained package 1, since three or more different kinds of soft foods 31 can be disposed, the number of kinds of colors and tastes can be increased.

[0092] As has been described above, according to the soft food-contained package 1 and packaging container 11 of the present embodiment, it is possible to prevent spreading of a taste and/or color between the soft food 31 and the seasoning liquid 32, and to prevent crumbling of the soft food 31.

[0093] The present invention is not limited to the above-described embodiments. At the stage of practicing the invention, various modifications may be made without departing from the spirit of the invention. In addition, the embodiments may properly be combined and practiced, and advantageous effects by the combinations can be obtained. Further, the embodiments include various inventions, and various inventions may be derived by properly combining structural elements selected from the disclosed structural elements. For example, even if some structural elements are omitted from all the structural elements disclosed in the embodiments, structures from which these structural elements are omitted may be derived as inventions in the case where the problem can be solved and the advantageous effects can be obtained.

Claims

1. A packaging container in which a soft food and a liquid are contained and which is sealed by a cover, the packaging container comprising a receptacle portion in which a recess portion, in which the soft food is contained, and a head space that is filled with the liquid above the soft food, are formed.
2. The packaging container of claim 1, wherein a plurality of the recess portions are provided.
3. The packaging container of claim 2, wherein a number of the recess portions is three or more.
4. The packaging container of claim 2, wherein the receptacle portion includes a planar portion, and the recess portions are formed on the planar portion by being spaced apart.
5. The packaging container of claim 4, wherein in the planar portion, at least one of the recess portions is disposed in a direction along a plane direction of the planar portion.
6. A soft food-contained package comprising: a soft food; a liquid; a receptacle portion in which a recess portion, in which the soft food is contained, and a head space that is filled with the liquid above the soft food, are formed; and a cover sealing the receptacle portion.
7. The soft food-contained package of claim 6, wherein a plurality of the recess portions are provided.
8. The soft food-contained package of claim 7, wherein a number of the recess portions is three or more.
9. The soft food-contained package of claim 7, wherein the soft foods including different ingredients are contained in the recess portions.
10. The soft food-contained package of claim 7, wherein the receptacle portion includes a planar portion, and the recess portions are formed on the planar portion by being spaced apart.

- 11.** The soft food-contained package of claim 10, wherein in the planar portion, at least one of the recess portions is disposed in a direction along a plane direction of the planar portion.
- 12.** The soft food-contained package of claim 6, wherein a height of the soft food contained in the recess portion is higher than a height of the head space from an upper surface of the soft food to a lower surface of the cover.
- 13.** The soft food-contained package of claim 10, wherein a height of the soft food contained in the recess portions is higher than a height from a bottom portion of of the soft food to the planar portion.
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